

# Project Design Phase-II

## Technology Stack (Architecture & Stack)

|              |                                                             |
|--------------|-------------------------------------------------------------|
| Date         | _____                                                       |
| Team ID      | _____                                                       |
| Project Name | OptiCrop: Smart Agricultural Production Optimization Engine |

**Table-1: Components & Technologies**

| S.No | Component           | Description                 | Technology                       |
|------|---------------------|-----------------------------|----------------------------------|
| 1    | User Interface      | Web interface               | HTML, CSS, JavaScript, Bootstrap |
| 2    | Application Logic-1 | Authentication & validation | Python (Flask)                   |
| 3    | Application Logic-2 | Crop recommendation         | Scikit-learn                     |
| 4    | Application Logic-3 | Data preprocessing          | Pandas, NumPy                    |
| 5    | Database            | User & crop data            | MySQL                            |
| 6    | Cloud Database      | Cloud storage               | Firebase/MongoDB Atlas           |
| 7    | File Storage        | Reports & datasets          | Local File System                |
| 8    | External API-1      | Weather                     | OpenWeatherMap API               |
| 9    | External API-2      | Soil info                   | SoilGrids API                    |
| 10   | ML Model            | Crop prediction             | Random Forest                    |
| 11   | Infrastructure      | Deployment                  | Render/AWS/Local                 |

**Table-2: Application Characteristics**

| S.No | Characteristic        | Description                       | Technology                         |
|------|-----------------------|-----------------------------------|------------------------------------|
| 1    | Open-Source           | Frameworks used                   | Flask, Pandas, NumPy, Scikit-learn |
| 2    | Security              | Authentication & password hashing | HTTPS, Werkzeug                    |
| 3    | Scalable Architecture | Three-tier architecture           | Flask MVC                          |
| 4    | Availability          | Web deployment                    | Render/AWS                         |
| 5    | Performance           | Fast ML prediction                | Scikit-learn                       |